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Phase behavior of systems with high CO_2 content: Experiments and thermodynamic modeling

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ABSTRACT

Pre-salt reservoirs are located in ultra-deep Brazilian waters. The petroleum production from these reservoirs is characterized by a high gas-oil ratio (GOR) and high content of CO_2 in the associated gas. The challenges for production in such the pre-salt reservoir conditions include understanding the unusual phase behavior observed in the laboratory from field samples. In this paper, we discuss the phase equilibrium data and thermodynamic modeling of the systems containing CO_2/n-$C_{16}H_{34}$, CH_4/CO_2/n-$C_{16}H_{34}$ and CH_4/CO_2/n-$C_{16}H_{34}$/$C_{14}H_{10}$. Transitions observed during this study were liquid-liquid (LL), vapor-liquid (VL), and vapor-liquid-liquid (VLL). The phase behavior investigated includes equilibrium conditions observed at temperatures from 293.1 K to 353.1 K and at pressures up to 58 MPa. The Peng-Robinson equation of state with classical van der Waals mixing rule was used to correlate the obtained experimental data. Results show a complex phase behavior for the binary system CO_2/n-$C_{16}H_{34}$ at 298 K, featuring two liquid-liquid regions and barotropic inversion near the VLL transition. The addition of methane to the previous system (1/1 mol ratio relative to CO_2) induces a phase behavior alteration in which the liquid-liquid transition disappears. The addition of a polyaromatic component (phenanthrene), whose influence over the mean molecular weight of the mixture under test is little, increased the saturation pressure. This study contributes to show that the use of a model representing the live-oil is a good strategy to understand how chemical and physical-chemical characteristics can induce complex phase behaviors, such as those observed in reservoirs with high carbon dioxide content.

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1. Introduction

Petroleum production design strongly depends on the phase behavior knowledge from reservoir fluids. As petroleum exploration moves towards deeper waters, the complexity of the fluids increases. As such, the investigation of flow assurance phenomena like precipitation and deposition of asphaltenes and waxes demand the development of suitable experimental setups and thermodynamic models capable of accounting for complex phase behavior [1]. Fluids with high gas-oil ratio (GOR) and high carbon dioxide (CO_2) content display such a phase behavior [2]. Brazilian Pre-salt fluids are just that.

Pre-salt reserves are the most significant discoveries in Brazilian territory in the last decades. The initial GOR in the Pre-salt fields is above 200 Sm^3/m [4,33], and the high content of CO_2 is a significant challenge for both producing and processing the fluids.

To assess such complex behavior, a research strategy is to use model fluids with well-defined compositions to understand the underlying phenomena that govern the phase behavior of the original oil. In this paper, fluid mixtures of methane (CH_4), carbon dioxide (CO_2) and n-hexadecane (n-$C_{16}H_{34}$) are considered as model systems for reservoir fluids. There are three governing factors behind the compositions chosen for this study: i) Live-oils have a large amount of methane, which governs the bubble point because it is the lighter component [5,6]. ii) The residual liquid phase at atmospheric condition has the typical molecular weight around 220–260 g/mol [5,7–10]. iii) Pre-salt fluids are rich in CO_2.

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Table 1
Chemical used in this work.

Chemical Name	Chemical Formula	Source	Mole fraction purity
Methane	CH_4	White Martins	0.995
Carbon dioxide	CO_2	White Martins	0.998
n-Hexadecane	n-$C_{16}H_{34}$	Sigma-Aldrich	0.990
Phenanthrene	$C_{14}H_{10}$	Sigma-Aldrich	0.980

In the literature, few studies investigate petroleum fluids with high CO_2 content [11,12]. These studies deal mainly with phase behavior of systems containing CO_2 and dead oil for enhanced oil recovery purpose [8,13–19]. Additionally, few studies [20] include the phase behavior of oil fraction with relatively high methane content. On the other hand, many experiments and modeling of phase equilibria covering binary systems with CO_2 and hydrocarbons [21–55] have been published along the years. Some authors reported barotropic inversion in systems of high CO_2 contents and hydrocarbons [33,37,56,57]. Studies with mixtures containing a higher number of alkane components and CO_2 are scarce [32,41,55,58,59]. Finally, the few available experiments/modeling of phase equilibrium of systems containing methane plus heavier alkanes are limited to a small range of pressures and temperatures [21,34,60,61]. The phase equilibrium of systems with methane and phenanthrene ($C_{14}H_{10}$) was investigated by Floter et al. [62] The available phase equilibrium data for ternary mixtures containing CO_2 are generally limited to systems with low carbon dioxide content [63,64].

This paper aims to investigate the phase behavior of model systems with high carbon dioxide and high methane content. n-Hexadecane is used as the representative molecule of the residual liquid, whereas the amount of carbon dioxide and methane, representing the associated gas, is varied. The influence of aromaticity is also investigated by adding phenanthrene to the ternary system. Experimental results are correlated with the Peng-Robinson equation of state using the classical van der Waals mixing rule.

2. Experimental

2.1. Chemicals

Table 1 shows the chemicals used in this study and their mole fraction purity. No further purification was employed.

2.2. Experimental methodology

The experimental apparatus consists of a high-pressure stainless cell with an effective maximum volume of 295 mL (Sanchez Technologies). The cell is equipped with a sapphire window, which allows the visualization of its content through the use of a digital microscope, allowing high-resolution real-time monitoring. The cell has an internal piston that enables one to control the sample pressure by changing the interior volume. Internal pressure and temperature are continuously monitored using, respectively, a pressure transducer (0–100 MPa) and a PT100 thermocouple. The cell is designed to reach temperatures up to 473.1 K and pressures up to 60 MPa. A magnetic stirrer is placed in the cell and the system positioned over stirring equipment to assure the mixture is homogeneous at all times.

The static synthetic method was used to obtain the phase equilibrium data. In such a procedure, a mixture with a known global composition is prepared. The addition of the chemicals was done using the gravimetric method. The conditions of pressure and temperature are then varied until the visual observation of phase transition through the digital microscope.

Prior to any experiment, n-pentane was used to clean the cell that was then evacuated. The liquid with the desired composition (n-hexadecane and eventually phenanthrene) was prepared by weighting each component and mixing them together. The experimental apparatus and the feed into the cell were described in detail in the previous work [65]. A small flask with the resulting mixture was placed on a high precision balance (Sartorius, 0.1 mg) and a capillary tube was carefully placed in the liquid and attached to the cell. A specific amount of liquid was then introduced in the high-pressure cell by opening the valve. As the cell was evacuated, the liquid flows in naturally. For typical experiments, the liquid mass introduced into the cell varied from 30 to 50 g. Once the liquid was in the cell, the gases were introduced using a similar strategy. Each gas component (CO_2 and CH_4) was separately held in a carbon fiber cylinder. The cylinder (empty weight 700 g) was placed on a high precision balance (Sartorius, 1 mg) and the initial mass was

Table 3
Properties of pure compounds considered in this work.

Component	P_c/MPa	T_c/K	ω
CO_2	7.383	304.21	0.2236
CH_4	4.599	190.56	0.0115
n-$C_{16}H_{34}$	1.4	723.00	0.7174
$C_{14}H_{10}$	2.9	864.00	0.4707

Table 4
Binary interaction coefficients k_{ij} ($l_{ij} = 0.0690$ between CO_2(1)/n-$C_{16}H_{34}$(2)).

Component	CH_4	CO_2	n-$C_{16}H_{34}$	$C_{14}H_{10}$
CH_4	–	0.0884	0.0661	0.2025
CO_2	–	–	0.0898	0.1080
n-$C_{16}H_{34}$	–	–	–	0.0039

Table 2
Systems investigated and P-T-x conditions used in this work.

Systems	x/mole fraction	T/K	P/MPa
CO_2 (1)/n-$C_{16}H_{34}$ (2)	$x_1 = 0.7793$; $x_1 = 0.8247$; $x_1 = 0.8798$; $x_1 = 0.9098$; $x_1 = 0.9369$; $x_1 = 0.9392$; $x_1 = 0.9570$; $x_1 = 0.9839$	293.5–353.1	5.81–55.86
CO_2 (1)/n-$C_{16}H_{34}$ (2)/CH_4 (3)	$x_1 = 0.3198$, $x_3 = 0.3201$; $x_1 = 0.3737$, $x_3 = 0.3749$; $x_1 = 0.4125$, $x_3 = 0.4139$	297.9–353.8	15.90–35.75
CO_2 (1)/n-$C_{16}H_{34}$ (2)/CH_4 (3)/$C_{14}H_{10}$ (4)	$x_1 = 0.4110$, $x_2 = 0.1624$, $x_3 = 0.4086$; $x_1 = 0.4096$, $x_2 = 0.1640$, $x_3 = 0.4100$; $x_1 = 0.3704$, $x_2 = 0.2385$, x = 0.3672; $x_1 = 0.3158$, $x_2 = 0.3350$, $x_3 = 0.3156$; $x_1 = 0.2656$, $x_2 = 0.4231$, $x_3 = 0.2690$	297.9–353.2	11.41–37.82

$u(x_1) = u(x_2) = u(x_3) = u(x_4) = 0.01$ are standard uncertainties for mole fractions.

Table 5
Comparison of the transition pressure results for CO_2 (1)/n-$C_{16}H_{34}$ (2) system, based on a mole fraction basis obtained for two experiments repeated (Exp. (1) and Exp. (2)).

		Exp. (1)[a]		Exp. (2)[b]			
		T/K	P/MPa	T/K	P/MPa/	|ΔT|/K	|ΔP|/MPa
VLE		312.7	16.56	313.5	16.40	0.8	0.16
		322.8	16.92	322.7	16.81	0.2	0.11
		332.9	18.15	332.6	18.04	0.2	0.11
		343.1	19.64	342.9	19.57	0.2	0.07
		352.8	21.06	352.6	21.00	0.2	0.06
VLLE	303.1	7.08	302.8	7.13	0.3	0.05	

[a] Mole fraction $x_1 = 0.9392$.
[b] mole fraction $x_1 = 0.9369$. $u(P) = 0.05$ MPa and $u(T) = 0.1$ K are standard uncertainties for pressure and temperature, respectively.

Table 6
Comparison of the transition pressure results for CO_2 (1)/CH_4 (2)/n-$C_{16}H_{34}$ (3)/$C_{14}H_{10}$ (4) system, based on a mole obtained for two experiments repeated Exp. (3) and Exp. (4).

	Exp. (3)[a]		Exp. (4)[b]			
	T/K	*P/MPa*	*T/K*	*P/MPa*	*|ΔT|/K*	*|ΔP|/MPa*
VLE	298.0	36.82	297.9	36.93	0.1	0.11
	302.9	36.67	303.1	36.89	0.1	0.23
	313.0	36.72	313.0	36.94	0.0	0.22
	323.0	36.95	323.0	37.13	0.0	0.18
	333.1	37.24	333.0	37.38	0.1	0.14
	343.2	37.56	343.2	37.66	0.0	0.10
	353.2	37.82	353.0	37.91	0.1	0.09

[a] Mol fraction $x_1 = 0.4096$, $x_2 = 0.1624$, $x_3 = 0.4100$ and $x_4 = 0.0180$.
[b] mole fraction $x_1 = 0.4110$, $x_2 = 0.1624$, $x_3 = 0.4086$ and $x_4 = 0.0150$. $u(P) = 0.05$ MPa and $u(T) = 0.1$ K are standard uncertainties for pressure and temperature transducers, respectively.

recorded. The cylinder was connected to the cell and the feeding valve was then opened. The gas flowed to the cell by pressure difference until a given mass previously calculated was introduced into the cell. Typical mass of gas sample varied from 5 to 15 g for CO_2 and of 2–6 g for CH_4. Through this procedure, the standard uncertainties in introduced mass were estimated to be less than 0.01 g for liquid mixture and 0.005 g for CO_2 and CH_4. For phase transition measurements, the temperature was set using an oven with rigorous temperature control (precision of 0.1 K). Once the desired temperature was reached, the sample was continuously mixed and the cell was pressurized until a homogeneous monophasic mixture is observed. While the temperature is kept constant, the pressure was then reduced in the rate of 0.1 MPa per minute until the observation of an incipient phase.

The model of live-oil used here was a mixture of n-$C_{16}H_{34}$ or n-$C_{16}H_{34}$/$C_{14}H_{10}$ and CO_2/CH_4. The compositions were based on Pre-salt oils data. The range of GOR from 100 to 450 Sm^3/Sm [3] and the similar mole ratio between methane and carbon dioxide characterized the target compositions. These GOR values were considered as a basis for estimating the mole fraction of gas and liquid phases using the equation of state. Additionally, the molar weight of the liquid phase (n-hexadecane or n-hexadecane + phenanthrene) is similar to the ave rage molar weight of dead oil reported by literature. Table 2 shows all the mixtures studied in this work, their respective mole fraction (x), as well as, the pressure (P) and the temperature (K) ranges investigated.

3. Thermodynamic modeling

Vapor-liquid equilibrium (VL), liquid-liquid equilibrium (LL) and vapor-liquid-liquid equilibrium (VLL) calculations are based on the isofugacity criteria, as shown in Eq. (1).

$$\hat{f}_i^{LI}\left(T,P,\underline{x}^{LI}\right)=\hat{f}_i^{LII}\left(T,P,\underline{x}^{LII}\right)=\hat{f}_i^{V}\left(T,P,\underline{x}^{V}\right) \quad i=1,2,\ldots,c \tag{1}$$

where $\underline{x}^V$ and $\underline{x}^L$ are the composition vectors of the vapor and liquid phases, respectively. T and P at equilibrium are temperature and pressure.

The fugacity coefficients of component i in the liquid phases (LI and LII) and vapor phases are obtained from the Peng-Robinson EoS (Equation (2)) with the classical mixing rules [66].

$$P=\frac{RT}{V_m-b}-\frac{a}{(V_m+b)(V_m-b)} \tag{2}$$

The classical mixing rule is described by Eq. (3) and Eq. (4):

$$a=\sum_i\sum_j x_i x_j\sqrt{a_i a_j}(1-k_{ij}) \tag{3}$$

$$b=\sum_i\sum_j x_i x_j\frac{(b_i+b_j)}{2}(1-l_{ij}) \tag{4}$$

The equation parameters, a and b, related with intermolecular attraction force and volume of molecules, respectively, are calculated by Eq. (5) and Eq. (6).

$$a_i=0.45724\frac{R^2T_{ci}^2}{P_{ci}}\left(1+k_i\left[1-\sqrt{\frac{T}{T_{ci}}}\right]\right)^2 \tag{5}$$

$$b_i=0.07780\frac{RT_{ci}}{P_{ci}} \tag{6}$$

The characteristic constant, k_i, related to the ω_i acentric factor of component i, is described by Eq. (7) and Eq. (8).

$$k_i=0.3764+1.5423\omega_i-0.2699\omega_i^2 \qquad \omega_i\le 0.49 \tag{7}$$

$$k_i=0.3764+1.4850\omega_i-0.1644\omega_i^2+0.01667\omega_i^3 \qquad \omega_i>0.49 \tag{8}$$

In the equations above, P_{ci} is the critical pressure and T_{ci} is the critical temperature of component i, T is the temperature, R is the universal gas constant, and V_m is the molar volume of the mixture.

The properties of pure compounds used in this study were taken from the literature [67] and are summarized in Table 3.

3.1. Binary interaction parameters calculation

The obtained k_{ij} parameters show a good adjustment for the vapor-liquid curve in the ternary and quaternary systems. Table 4 summarizes all the binary interaction parameters used in this work.

Table 7
Experimental phase equilibrium data for the CO_2 (1)/n-$C_{16}H_{34}$ (2) system.

L − LL transitions		LL − VLL transitions		L − VL transitions	
T/K	*P*/MPa	*T*/K	*P*/MPa	*T*/K	*P*/MPa
$x_1 = 0.9838$					
297.9	10.22	294.4	5.81	303.0	10.27
299.0	10.19	298.0	6.33	313.2	12.12
		302.6	7.01	323.1	14.12
				333.1	15.99
				343.5	17.78
				352.8	19.27
$x_1 = 0.9570$					
297.6	25.08	294.3	5.83	303.4	17.43
298.2	23.83	297.9	6.39	313.0	15.83
		302.8	7.10	322.9	16.53
				332.8	17.91
				342.8	19.40
				352.8	20.91
$x_1 = 0.9370$					
298.0	46.70	293.7	5.81	302.9	19.22
298.1	46.95	297.8	6.37	313.5	16.40
298.9	54.26	302.8	7.12	322.7	16.81
299.0	55.81			332.6	18.04
297.7	34.74			342.9	19.57
298.1	29.97			352.6	20.99
298.4	28.37			363.0	22.47
298.6	27.13				
$x_1 = 0.9090$					
298.6	49.47	293.6	5.83	303.1	19.97
299.1	58.43	297.9	6.41	313.0	16.64
298.5	48.73	302.6	7.10	322.7	16.85
298.6	49.14			333.1	17.41
298.5	48.44			342.9	19.51
298.5	48.71			352.9	21.01
298.7	50.55			363.4	22.44
298.1	34.14				
298.9	28.37				
298.2	33.31				
298.1	33.63				
298.2	33.03				
298.3	31.65				
$x_1 = 0.8798$					
298.0	49.11	293.7	5.91	303.4	18.91
298.0	47.80	297.9	6.49	313.3	16.31
298.3	50.97	303.3	7.27	323.6	16.74
298.8	55.86			333.6	17.91
297.7	34.12			343.5	19.31
297.8	32.18			352.9	20.65
298.0	30.51				
298.4	28.13				
$x_1 = 0.8247$					
297.8	17.05	293.8	5.85	303.3	13.53
298.2	16.57	298.3	6.56	313.0	13.05
		302.7	7.21	323.0	13.96
				332.9	15.50
				343.1	17.10
				352.7	18.27
$x_1 = 0.7793$					
297.8	10.13	294.1	5.85	303.2	9.13
298.8	9.72	297.7	6.35	313.0	10.17
		303.2	7.26	323.0	12.00
				332.9	13.67
				343.0	15.28
				353.0	16.81

$u(P) = 0.05$ MPa and $u(T) = 0.1$ K are standard uncertainties for pressure and temperature transducers, respectively.

The following objective function was adopted:

$$OF = min \sum_{i=1}^{Nexp} \left| \frac{P_i^{calc} - P_i^{exp}}{P_i^{exp}} \right| \quad (9)$$

To correlate the experimental phase transition data, a bubble point calculation algorithm, based on successive substitution and Newton methods, was developed in Fortran 90. The algorithm also includes a stability analysis which uses the criterion of tangent plan distance [68]. For the system CO_2 (1)/n-$C_{16}H_{34}$ (2), the binary interaction parameters were estimated by using the Hook-Jeeves/SIMPLEX algorithm [69,70] through fitting the experimental VLE and LLE data from Table 7. Since liquid-liquid transition is observed only for the binary system CO_2 (1)/n-$C_{16}H_{34}$ (2), the l_{ij} for this system is calculated. For all other binary systems, the l_{ij} is set to

Table 8
Experimental phase equilibrium data for the CO_2 (1)/n-$C_{16}H_{34}$ (2)/CH_4 (3) system.

L - VL					
$x_1 = 0.4125$		$x_1 = 0.3737$		$x_1 = 0.3198$	
$x_2 = 0.1736$		$x_2 = 0.2514$		$x_2 = 0.3601$	
$x_3 = 0.4139$		$x_3 = 0.3749$		$x_3 = 0.3201$	
T/K	*P/MPa*	*T/K*	*P/MPa*	*T/K*	*P/MPa*
298.0	33.57	297.9	24.67	299.3	15.90
303.0	33.64	303.6	25.17	303.3	16.32
313.3	34.00	313.1	26.01	313.2	17.17
323.1	34.45	322.9	26.76	323.1	18.06
333.5	34.93	333.8	27.57	332.9	18.84
343.0	35.35	342.8	28.19	343.0	19.57
352.8	35.75	352.9	28.80	353.0	20.20

u(P) = 0.05 MPa and u(T) = 0.1 K are standard uncertainties for pressure and temperature transducers, respectively.

Table 9
Experimental phase equilibrium data for the CO_2 (1)/n-$C_{16}H_{34}$ (2)/CH_4 (3)/$C_{14}H_{10}$ (4) system.

L - VL							
$x_1 = 0.2656$		$x_1 = 0.3158$		$x_1 = 0.3704$		$x_1 = 0.4096$	
$x_2 = 0.4231$		$x_2 = 0.3350$		$x_2 = 0.2385$		$x_2 = 0.1640$	
$x_3 = 0.2690$		$x_3 = 0.3156$		$x_3 = 0.3672$		$x_3 = 0.4100$	
$x_4 = 0.0423$		*$x_4 = 0.0336$*		*$x_4 = 0.0239$*		*$x_4 = 0.0164$*	
T/K	*P/MPa*	*T/K*	*P/MPa*	*T/K*	*P/MPa*	*T/K*	*P/MPa*
298.0	11.41	297.9	26.51	298.0	16.74	298.0	36.81
303.1	11.82	304.1	26.97	302.9	17.30	300.2	36.70
313.1	12.55	308.1	27.24	308.5	17.94	302.9	36.67
323.2	13.23	312.9	27.54	312.9	18.33	305.2	36.66
333.2	13.86	323.5	28.19	323.1	19.27	308.0	36.66
343.0	14.45	333.0	28.83	333.3	20.03	310.2	36.68
353.1	15.03	343.4	29.44	343.1	20.69	313.0	36.72
		353.0	29.95	353.1	21.33	323.0	36.95
						333.1	37.24
						343.2	37.56
						353.2	37.82

u(P) = 0.05 MPa and u(T) = 0.1 K are standard uncertainties for pressure and temperature transducers, respectively.

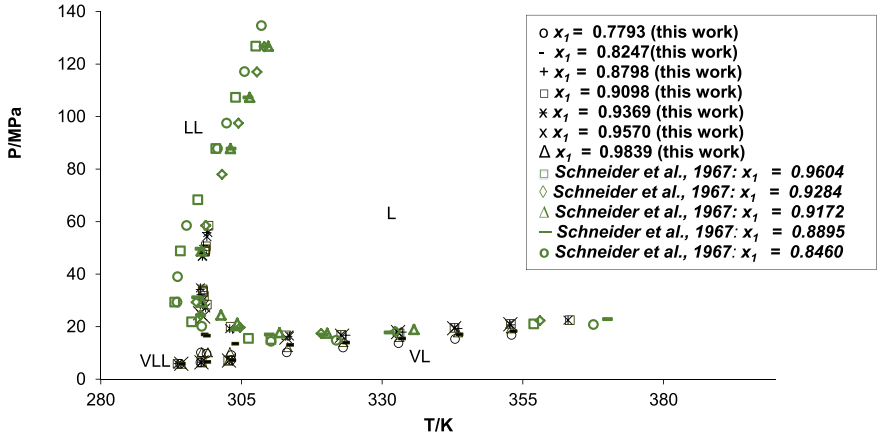


Fig. 1. Comparison between experimental data (this work) and literature data [71] for the CO_2(1)/n-$C_{16}H_{34}$(2) system.

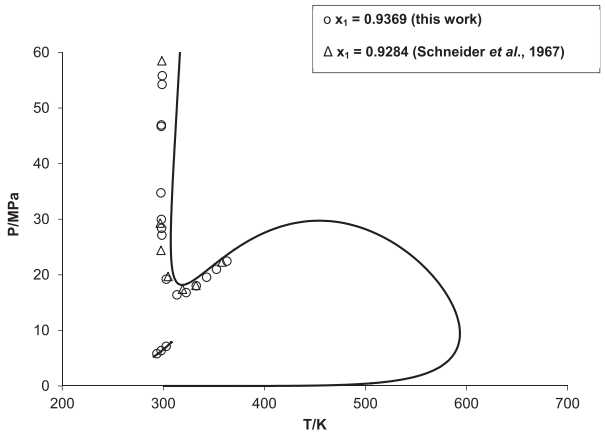


Fig. 2. Comparison between experimental and literature data for the CO_2(1)/n-$C_{16}H_{34}$(2) system: (o) $x_1 = 0.9369$ (this work) and (Δ) $x_1 = 0.9284$[71]; —— PR.

zero. For the ternary system, the binary k_{ij} and l_{ij} parameters for the system CO_2 (1)/n-$C_{16}H_{34}$ (2) previously estimated were kept constant, and the k_{ij} parameters for the systems CO_2 (1)/CH_4 (3) and n-$C_{16}H_{34}$ (2)/CH_4 (3) were estimated fitting VLE data from Table 8. The k_{ij} parameters already obtained were fixed for the quaternary system and the k_{ij} parameters for CO_2 (1)/$C_{14}H_{10}$ (4), CH_4 (3)/$C_{14}H_{10}$ (4) and n-$C_{16}H_{34}$ (2)/$C_{14}H_{10}$ (4) systems were calculated with VLE data from Table 9.

4. Results and discussion

4.1. Reproducibility

In order to check the reproducibility of the experimental unit, we performed VL and VLL equilibrium experiments for binary mixtures of carbon dioxide and n-hexadecane, and quaternary systems containing methane, carbon dioxide, phenanthrene, and n-hexadecane. For each set of experiments, the entire cell is assembled, and the desired mixture is prepared. Results in Table 5 are those of binary system, whose experiments are Exp. (1) and Exp. (2). In Table 6, the results for the quaternary system (Exp. (3) and Exp. (4)) are shown. Good reproducibility was found, indicating the experiments set up are validated.

4.2. CO_2/n-$C_{16}H_{34}$ system

Table 5 shows the experimental data for the CO_2/n-$C_{16}H_{34}$ system. Figs. 1 and 2 show a comparison between measured values and those reported in the literature [71]. As one can observe in Fig. 3, the experimental data obtained in this work are consistent with data from the literature, including VL, LL, and VLL phase transitions. Figs. 2 and 3 also show the results using the Peng-Robinson EoS.

Fig. 4 reports the P-T diagrams for several CO_2 concentrations. The CO_2/n-$C_{16}H_{34}$ system exhibits a type-III phase behavior, according to Konynenburg-Scott classification [74]. Type-III systems are characterized by the presence of an upper critical end-point (UCEP) at temperatures close to the critical temperature of the

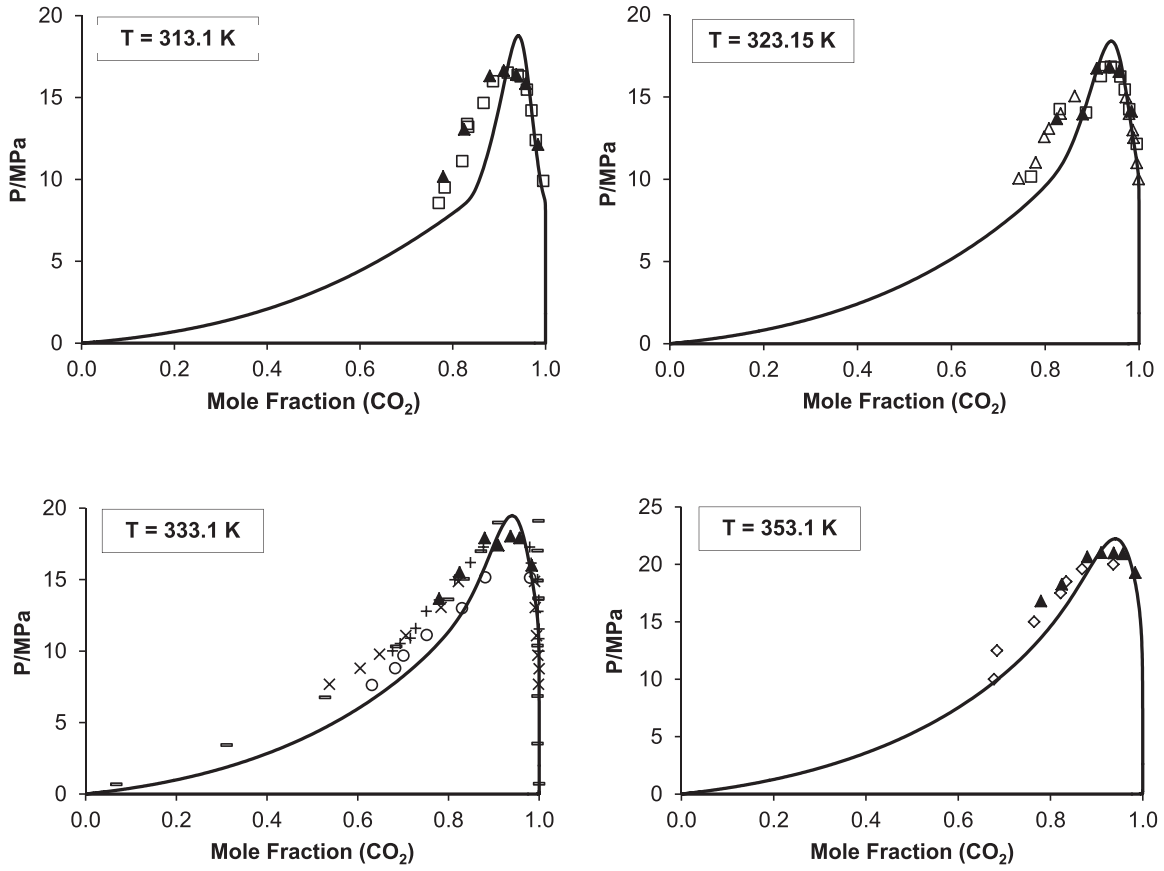


Fig. 3. Comparison among modeling, data obtained in this work, and literature data. (□) [41], (Δ) [72], (○) [73], (x) [27], (+) [28], (−) [25], (◊) [35]; (▲) this work; (—)PR, in CO_2(1)/n-$C_{16}H_{34}$(2) system — P-x diagrams.

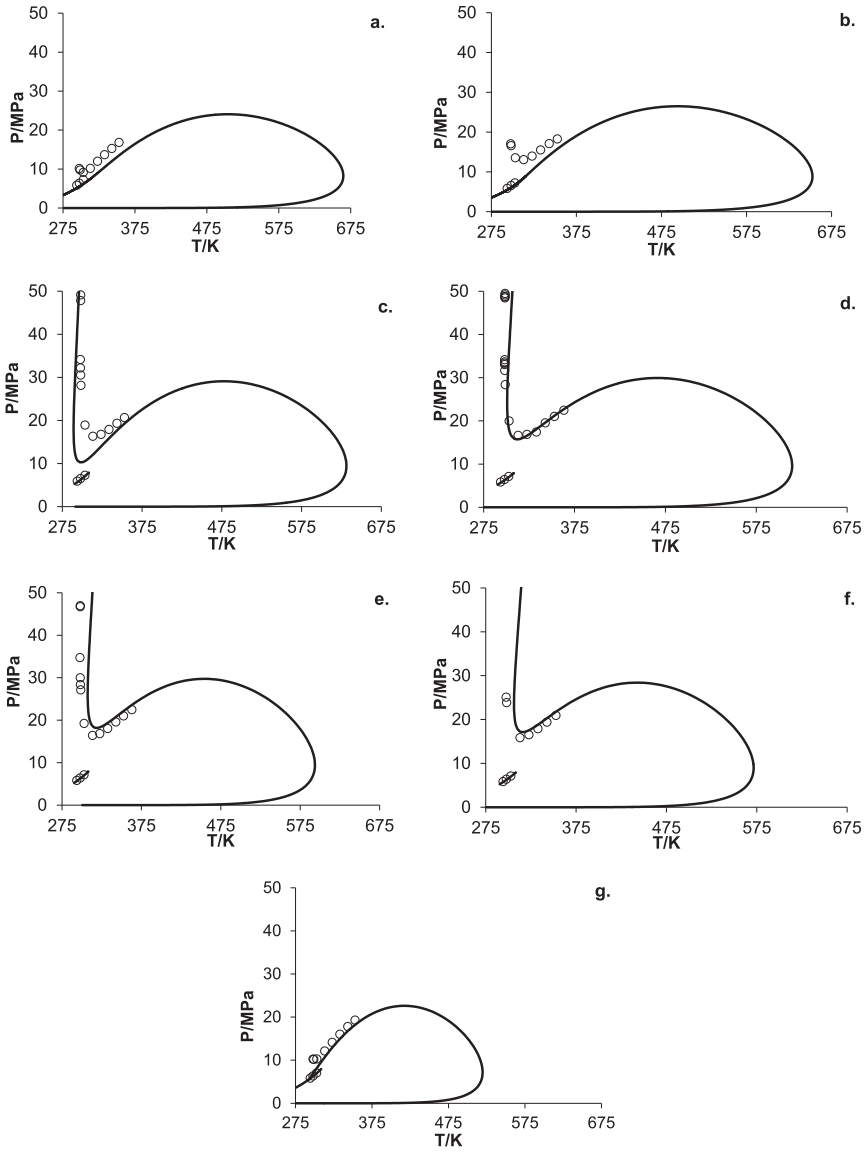


Fig. 4. P-T diagrams of the CO_2(1)/n-$C_{16}H_{34}$(2) system (○) exp. data, **a**. $x_1 = 0.7793$; **b**. $x_1 = 0.8247$; **c**. $x_1 = 0.8798$; **d**. $x_1 = 0.9098$; **e**. $x_1 = 0.9369$; **f**. $x_1 = 0.9570$; **g**. $x_1 = 0.9839$; — PR.

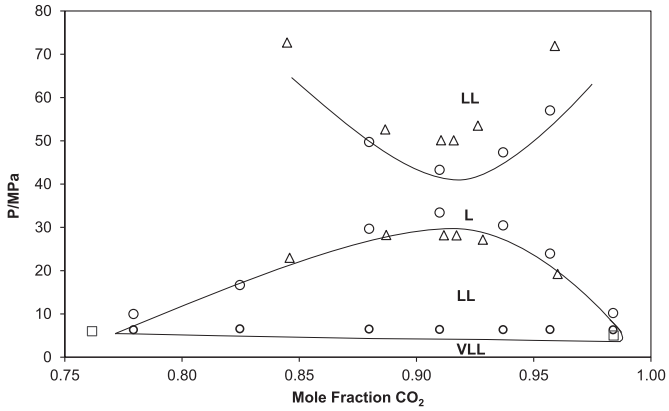


Fig. 5. P-x diagram of CO_2 (1)/n-$C_{16}H_{34}$ (2) system at 298 K (○), — The lines are to guide the eye. (Δ) [71] and (□) [75].

lighter component (CO_2). The Peng-Robinson EOS shows an excellent performance in describing the experimental data.

The CO_2/n-$C_{16}H_{34}$ system displays an interesting phase behavior known as the miscibility window at 298.1 K. Considering the P-x diagram, the phase behavior with high CO_2 content shows immiscibility between two high-density phases at higher and lower pressures. There is an intermediate pressure range showing total miscibility. Fig. 5 shows the data obtained in this work and those from Schneider (1967) [71] and Francis (1954) [73]. The pressure transition values of the LL immiscibility regions are very sensitive to temperature variation. As such, these data (298.1 K) were obtained as the average of four different measurements. The experimental results are shown in Table 7 (L-LL transitions at high and low pressures).

Orr Jr. and Jensen (1984) [14] found a similar behavior, but their mixtures were composed by CO_2/crude oil at 305 K, 314 K, and 322 K. The three phases (VLL) were found for mixtures with CO_2 content higher than 70 mol % at 314 K. Cardoso et al. (2015) [2]

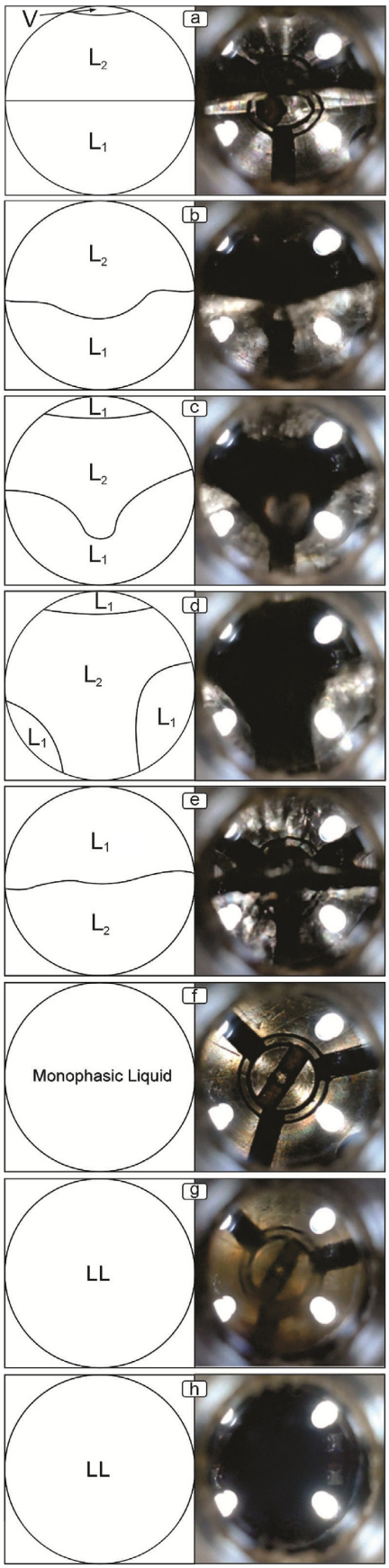


Fig. 6. Isothermal pressurization of CO_2 (1)/n-$C_{16}H_{34}$ (2) at 298.1 K ($x_1 = 0.9098$). In the left column, there a sequence of illustrative images of real images located in the right column – a. At pressure about 6 MPa the system presents VLL, where liquid phase below is named by L_1 and phase above, by L_2; b., c. and d. At constant pressurization, the system presents the phenomena of phase inversion, where liquid L1 has been changing the position with L_2; e. At pressure about 20 MPa, the system shows two liquid phases; however, L_1 is above L_2; f. At pressure about 40 MPa, the system presents a monophasic liquid; g. At pressure above 40 MPa, there is a LL transition with two phases of similar densities; h. At pressure above 55 MPa shows two liquid phases, which do not separate.

reported an unusual phase behavior for CO_2 rich live-oil samples from the Pre-salt reservoirs, where two high-density phases coexist at reservoir conditions or appear during the isothermal depressurization. Although the thermodynamic conditions are not equal, the results and data reported here provide an insight for a better understanding of the region of one- and two-liquid phases.

Another phenomenon is observed during isothermal pressurization with 90.98 mol % CO_2 content at 298.1 K: The mass barotropic inversion (density inversion). Charoensombut-Amon et al. (1986) [25] have observed similar behavior for this system at 14.14 MPa and 308.1 K van der Steen et al. (1989) [29] have reported this behavior for n-$C_{16}H_{34}$ and other kinds of paraffin with CO_2. Fig. 6 shows the sequence of images during isothermal pressurization around 298.1 K, where one can observe the barotropic inversion followed by the miscibility region. Image A presents VLL equilibria, at pressures below 6.3 MPa. By increasing the pressure of the system, the phenomenon of barotropic inversion is observed, where the liquid below (L_1) is richer in CO_2. With the continuous increase of pressure, it is possible to find two liquid phases (LL), in Image E, with L_1 above L_2. The system becomes monophasic at 40 MPa (Image F) (miscibility window). At higher pressures (pressure above 50 MPa), there is another liquid to liquid-liquid transition, known as an upper critical transition (Image G). The volume of the second liquid phase increases, and the system becomes immiscible; consequently, it stays cloudy (Images G and H). These events can be observed in a P-x diagram (Fig. 5). A video (Video 01) containing the sequence of events is available in the Supporting Material.

Supplementary video related to this article can be found at https://doi.org/10.1016/j.fluid.2020.112574

4.3. CO_2/CH_4/n-$C_{16}H_{34}$ system

Table 8 shows the experimental data obtained for the CO_2/CH_4/n-$C_{16}H_{34}$ system. Only liquid-vapor transition was observed at a constant CO_2/CH_4 mole fraction ratio (equal to 1), which is common in some Pre-salt reservoirs. In Fig. 7, experimental data and thermodynamic modeling for this system are drawing. It is possible to note that the phase equilibrium data obtained in the present study are consistent with the literature [63,64] (Fig. 8).

4.4. CO_2/CH_4/n-$C_{16}H_{34}$/$C_{14}H_{10}$ system

Table 9 shows the experimental data obtained for the CO_2/n-$C_{16}H_{34}$/CH_4/$C_{14}H_{10}$ system. The corresponding P-T diagram is shown in Fig. 9. These experiments have a molar ratio of $C_{14}H_{10}$ to n-$C_{16}H_{34}$ fixed at 0.1. The Peng-Robinson model was described for experimental data from this work.

This system has average molecular mass similar to CO_2/n-$C_{16}H_{34}$/CH_4 system. The presence of phenanthrene (aromatic compounds) increases the saturation pressure, as observed in Fig. 10. For the higher gas concentration, this effect is pronounced.

5. Conclusions

We investigated the phase behavior of the CO_2/n-$C_{16}H_{34}$, CO_2/

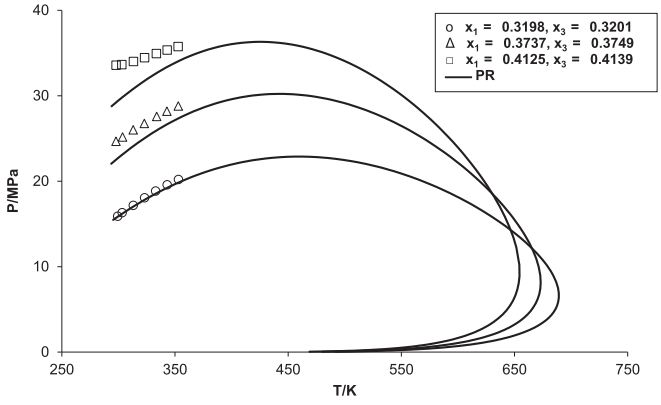


Fig. 7. Experimental data from this work for the CO_2 (1)/n-$C_{16}H_{34}$ (2)/CH_4 (3) system - P-T diagram.

CH_4/n-$C_{16}H_{34}$ and CO_2/CH_4/n-$C_{16}H_{34}$/$C_{14}H_{10}$ mixtures. The binary system CO_2/n-$C_{16}H_{34}$ shows a complex phase behavior with two liquid-liquid zones. Both CO_2/CH_4/n-$C_{16}H_{34}$ and CO_2/CH_4/n-$C_{16}H_{34}$/$C_{14}H_{10}$ systems show LCST transitions in the range of investigated conditions. Results indicate that the high CO_2 content in Pre-salt fluids could substantially influence the high-pressure phase behavior. The presence of polyaromatic compounds (such as phenanthrene) increases the saturation pressure of the mixture, without considerable change in the mean molecular mass. The Peng-Robinson equation showed good performance to correlate the experimental data with a constant binary interaction parameter. The approach of live-oil concept can be useful to better deal with the complexity of petroleum by using a well-characterized fluid. This study showed that the high-density phases equilibrium at reservoir conditions was reproduced neither by addition of methane oil nor by the addition of aromatic molecule. On the other hand, the binary mixture CO_2/n-$C_{16}H_{34}$ exhibits a similar behavior (298 K) at upper liquid-liquid equilibria. The complex phase behavior observed in reservoir suggests that a composition of live-oil containing more compounds can enhance the comprehension of the petroleum fluid behavior.

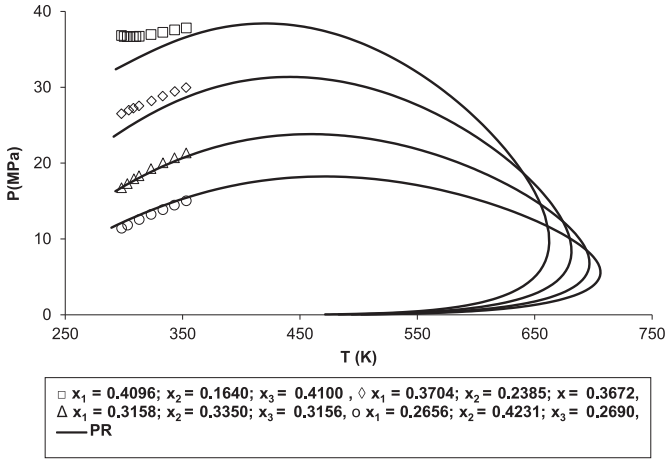


Fig. 9. Experimental data from this work for the CO_2 (1)/n-$C_{16}H_{34}$ (2)/CH_4 (3)/$C_{14}H_{10}$ (4) system − P-T diagram.

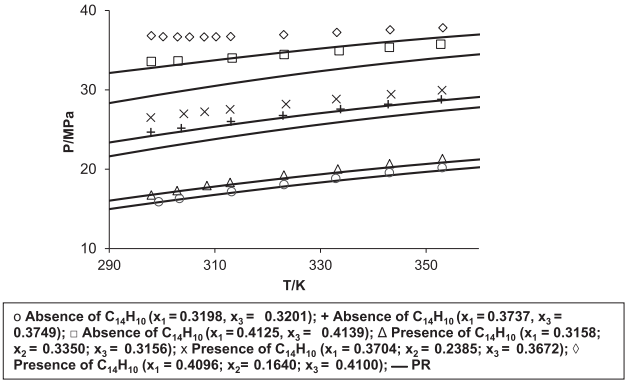


Fig. 10. Analysis of the effect of $C_{14}H_{10}$ (4) in CO_2 (1)/n-$C_{16}H_{34}$ (2)/CH_4 (3) system.

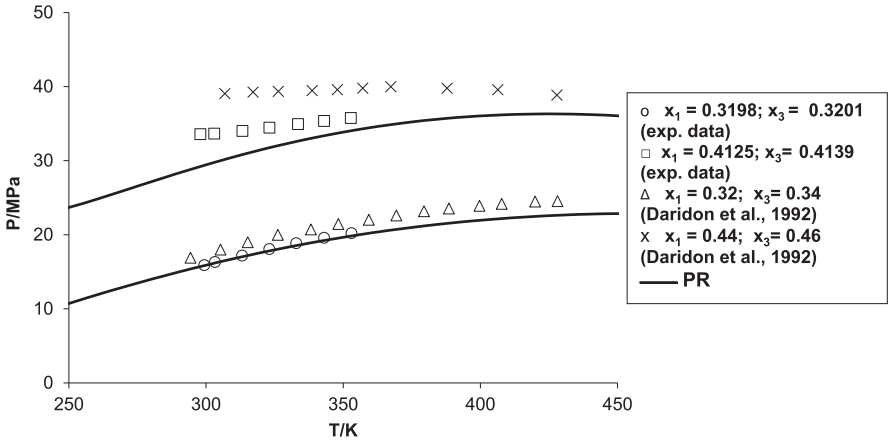


Fig. 8. Comparison with experimental data from this work and literature data [63] for the CO_2 (1)/n-$C_{16}H_{34}$ (2)/CH_4 (3) system − P-T diagram.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Ana P.P. Simoncelli: Data curation, Writing - original draft, Writing - original draft. **Wilfred Goméz:** Data curation, Software, Validation, Writing - original draft. **Rafael M. Charin:** Supervision, Investigation. **Felipe P. Fleming:** Conceptualization, Methodology, Writing - review & editing. **Papa M. Ndiaye:** Conceptualization, Supervision, Software. **Frederico W. Tavares:** Supervision, Conceptualization, Methodology, Writing - review & editing.

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List of symbols

c	total number of compounds
a	energy parameter
b	co-volume
T	temperature
P	pressure
V_m	molar volume
R	universal gas constant
x	mole fraction
f	fugacity
T_c	critical temperature
P_c	critical pressure
k_{ij}	parameter iteration binary term energy
l_{ij}	parameter iteration binary co-volume

Superscripts

L	liquid phase
V	vapor phase
Exp	experimental
calc	calculated

Subscripts

i, j	component index

Greek letters

ω	acentric factor
u	standard uncertainty

Abbreviations

EoS	equation of state
PR	Peng-Robinson
VL	vapor-liquid
LL	liquid-liquid
VLL	vapor-liquid-liquid

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.fluid.2020.112574.

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